THE ROLES OF GRAMMAR AND PRAGMATICS IN FOCUS DEPENDENCIES*

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1 Focus dependencies

In question focus (QF), illustrated in (1), the placement of focus in the answer corresponds to the wh-phrase in the question, an instance of so-called question-answer congruence. In free focus (FF), illustrated in (2), the placement of focus (here in a correction) corresponds to what has been changed in a sentence compared to previous material. And in association with focus (AF), illustrated in (3), the placement of focus co-varies with the entailments of a semantic operator (in this case, 'only', with the possible entailment in (a) but not in (b) that she did not give cake to Bob and the possible entailment in (b) but not in (a) that she did not give cookies to Bill).[1]

(1) Question focus (QF)
  a. (What did she give to Bob?) She gave $[\text{COOKIES}]_{\text{F}}$ to Bob
  b. (Who did she give cookies to?) She gave cookies to $[\text{BOB}]_{\text{F}}$

(2) Free focus (FF)
  a. (She gave candy to Bob.) No, she gave $[\text{COOKIES}]_{\text{F}}$ to Bob
  b. (She gave cookies to Bill.) No, she gave cookies to $[\text{BOB}]_{\text{F}}$

(3) Association with focus (AF)
  a. She **only** gave $[\text{COOKIES}]_{\text{F}}$ to Bob
  b. She **only** gave cookies to $[\text{BOB}]_{\text{F}}$

My goal here is to explore the possibility that such dependencies have a unified account in terms of the following pragmatic condition, which does not make reference to focus:

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[1] F-marking a constituent X, notated as $[X]_{\text{F}}$, leads to prosodic prominence within X, often by pitch accent, notated using allcaps. Semantically, F-marking X within ϕ is often taken to lead to a focus semantic value (FSV), $[\![\,\phi\,]\!]_{\text{f}}$, through allowable substitutions for X within ϕ. $[\![\,\phi\,]\!]_{\text{f}}$ is a second dimension of meaning, distinct from the ordinary semantic value $[\![\,\phi\,]\!]_{\text{o}}$. I will refer to FSVs in much of the discussion but eventually argue that they do not exist.

(4) For S to be felicitous in context C, it should be a good answer to a good question in C

(4) is often taken for granted in pragmatics: it is commonly accepted that sentences in discourse should be understood as answers to questions; given that, it stands to reason that the question and the answer should be good.[2] But it is not usually taken to suffice for focus dependencies such as those in (1)–(3), and the literature has adopted additional machinery, perhaps the best-known being Rooth (1992)'s anaphoric operator $\sim$. Syntactically, $\sim$ attaches to a silent variable v and then to a constituent X, resulting in the configuration $[\sim_{\mathrm{v}} X]$. Semantically, $\sim$ ensures that the value of v is a subset of $[\![\mathrm{X}]\!]_{\mathrm{f}}$. Rooth uses $\sim$ as the basis for a unified account of focus dependencies, and it has been used in many other, often more fragmentary accounts of focus since.

For QF in examples such as (1), congruence (matching F-marking in the answer against the wh-elements in the question) can be seen as an instantiation of (4) in terms of form. This shape requirement can be restated semantically, as a requirement that the FSV of S match the semantics of the question, a set of possible answers along the lines of Hamblin (1973) and Karttunen (1977). The semantic restatement can use $\sim$: $\sim_{\mathrm{v}}$ attaches to S, and the value of v is set to Q, the denotation of the question, ensuring that $Q \subseteq [\![\mathrm{S}]\!]_{\mathrm{f}}$. This requires a constituent dominating 'cookies' to be F-marked in (1a) and a constituent dominating 'Bob' to be F-marked in (1b). To prevent over-focusing (e.g., F-marking both 'cookies' and 'Bob'), we can add a condition of minimality, following Schwarzschild (1993, 1997).

(5) CONGRUENCE ($\sim$-sensitive):
   a. S is parsed as $[\sim_{\mathrm{v}} S]$; v is anaphoric to the denotation of the question that S addresses
   b. S is minimally F-marked to achieve (5a)

Note that there is a different instantiation of (4), in terms of content rather than form: for S to be a good answer to a question, it should be relevant to it. Relevance is a pragmatic requirement, and differently from congruence (which relied on the semantic notion of question as a set of possible answers) it concerns the pragmatic notion of question as a partition Π of the context set C.[3] S is relevant to Π exactly when its ordinary semantic value does not cut across cells in Π. That is, when $[\![\mathrm{S}]\!]_{\mathrm{o}} \cap C$ is a union of cells in Π.[4] In principle, congruence and relevance might both be real constraints, with relevance checked against the pragmatic question derived from the semantic question that congruence refers to. But one may also wonder whether one of the two constraints is reducible to the other, and below I will suggest that this is in fact the case and that the appearance of congruence reduces to relevance.

Before that, I will review some of the roles that $\sim$ has been taken to play in FF (§2) and AF (§3) and the motivation for adopting it. In both cases, I will show that $\sim$ is in fact a liability, doing

[2] Collingwood (1940:p. 23): "Every statement that anybody ever makes is made in answer to a question." Works that rely on something like (4) include Schwarzschild (1993), von Fintel (1994), Roberts (1996), Büring (2003), Beaver and Clark (2008), Krifka (2008), Katzir and Singh (2015), Büring (2019), Fox (2019), Bar-Lev and Fox (2023), Bar-Lev (2024), and Katzir (2024), among many others.

[3] A semantic question (a set of possible answers) can be mapped to a pragmatic question (that is, a partition) whose cells are the different consistent assignments of truth values to the elements of the semantic question. Fox (2019) notes significant implications of this mapping to the theory of questions, a matter that I will rely on below.

[4] This notion of relevance, from Lewis (1988), captures the idea of staying on topic. Other work, including Roberts (1996) and Büring (2003), replaces Lewis's requirement with a requirement of being informative (or otherwise helpful) with respect to a question. To the extent that the reduction below is successful, it supports Lewis's requirement.

damage whenever its predictions diverge from those of a theory of QF. I will then return to QF in §4 and suggest that there, too, $\sim$ does more harm than good. The emerging picture makes no use of focus anaphoricity. In fact, it has no room for FSVs within the grammar, leaving focus as a purely pragmatic notion. Challenges remain, and I will highlight some of them below.[5]

2 Free focus

In principle, one can try to account for FF by reducing it to QF. In (2), we can take each of the two examples as involving a reconstructed question, perhaps the same one as the corresponding example in (1): "What did she give to Bob?" for (2a) and "Who did she give cookies to?" for (2b). If these are the questions, the focus pattern follows, just as in (1). Roberts (1996) offers this kind of reduction. See also Büring 2019 and Katzir 2024, among others.

More commonly, however, FF has been analyzed in terms of an anaphoric mechanism that goes well beyond what was used for QF. Rephrasing Schwarzschild (1999):[6]

(6) Anaphoric FF:
  a. $\sim_v$ attaches at every node in *S* that is not F-marked
  b. *S* is minimally F-marked to achieve (6a)

In (2a), the difference between "She gave cookies to Bob" and the context-setting "She gave candy to Bob" arises from the different choice of direct object: 'cookies' vs. 'candy'. F-marking 'cookies' makes it possible to attach $\sim$ to all other nodes in the structure, with the F-marked 'cookies' effectively becoming a wildcard for anaphoric matching: [gave candy to Bob] can serve as an antecedent to [gave cookies$_F$ to Bob], etc. This is the minimal F-marking that allows all non-F-marked nodes to receive $\sim$, so this parse is chosen. Analogously for (2b).

The following might serve as informal motivation for this move. Reducing FF to QF using (4) and (5) requires talking about reconstructed questions, and this reconstruction comes uncomfortably close to mind reading.[7] It also requires stretching the notion of anaphoricity to material that is not explicitly provided in discourse. One might hope that with an anaphoric system such as (6) the account can be more mechanical, with no need to reason about speakers' minds.

Perhaps more importantly, the anaphoric approach has empirical motivation, based on apparent local anaphoric deaccenting (Schwarzschild, 1999):

(7) (John is rich and has many expensive convertibles. Guess what he brought to his niece's wedding . . .) He brought a CHEAP convertible

In (7), the contextual question is what John brought, which asks about the entire direct object and includes potential answers such as that John brought a pony. But accent falls narrowly on the adjective 'cheap', which seems puzzling for the reductive view of FF as QF: on that view, we expect F-marking on the entire direct object rather than just on the adjective. With F-marking on

[5]An earlier proposal for eliminating focus from within the grammar is outlined in Kadmon and Sevi (2011). Their approach, which is very different from the present one, relies on a proposed mapping between prosody and the predictability of linguistic material in context. I will not provide a detailed comparison with their approach here.

[6]Note that (6a) leads to $\sim_v$ attaching also to constituents with no F-marking. In such cases the FSV of the node is the singleton set of its ordinary semantic value, and similarly for the anaphoric antecedent of *v*.

[7]See Bolinger (1972) for an early defense of appealing to mind reading in an account of stress.

the adjective alone, $[\![\text{S}]\!]_{\text{f}}$ will only include elements that differ with respect to the adjective, such as that he brought an expensive convertible and that he brought a new convertible, but not elements such as that he brought a pony. So $Q \not\subseteq [\![\text{S}]\!]_{\text{f}}$, and congruence does not hold.

On the anaphoric (6), (7) is unproblematic. The minimal F-marking that allows all remaining nodes to receive $\sim$ is on both the adjective 'cheap' and the entire direct object 'a cheap convertible'. Fewer F-marks would violate the anaphoric requirement of $\sim$, and any additional F-marks would violate minimality. This F-marking corresponds to the observed prosody. See Schwarzschild (1999) and much later work for discussion.

Building on Katzir (2024), I wish to make the following points. (7) does, in fact, have an account in terms of QF, and this account has welcome predictions that are not made by (6). The account admittedly relies on a certain amount of mind reading, but the literature provides reasons to think that this is inevitable even if one adopts (6).

The account of (7) in terms of QF relies on expectations. In the given context, an expected answer is that John brought an expensive convertible. If that had been the answer, the entire direct object would have been F-marked, corresponding to the question. But the actual answer is not the expected one, and in this context it is naturally understood as a correction of the expectation. If the expectation had been stated explicitly (e.g., "I guess he brought [an expensive convertible]$_{\text{F}}$"), the correction would have the same accent on 'cheap' that we find in (7), presumably with F-marking on both the adjective and the entire direct object. (7), then, suggests that discourse may proceed as if the expected answer had actually been given.

If apparent anaphoric deaccenting in (7) is due to an accommodated expectation, it should be possible to change the accent by manipulating expectations. For example, if there is a societal norm that people only bring cheap versions of things as presents (cheap books, cheap ponies, etc.) while expensive versions are frowned upon, then 'cheap' will not be accented in this context, but 'expensive' in "He brought an expensive convertible" will. This seems correct. See Kadmon and Sevi (2011) and Katzir (2024) for further evidence that (7) involves an accommodated expectation.

The current account predicts that F-marking follows the logic of questions and answers rather than that of anaphoricity. Evidence from Wagner (2006) suggests that this is indeed the case. If we change 'cheap' to 'red', the result is odd:

(8) (John is rich and has many expensive convertibles. Guess what he brought to his niece's wedding ...) #He brought a RED convertible

In terms of anaphoricity, nothing has changed between (7) and (8): as far as (6) is concerned, the two are entirely parallel. The oddness of accent on 'red' in (8) is therefore puzzling. In terms of questions and answers, on the other hand, there is a difference: the version with 'cheap' can successfully correct the expectation, but the version with 'red' cannot. This is due to properties of questions and answers, noted and discussed by Fox (2007, 2019), that allow "He brought an expensive convertible" and "He brought a cheap convertible" to answer the same question but prevent "He brought an expensive convertible" and "He brought a red convertible" from doing so. Corrections involve two answers to the same question, which is possible in (7) but not in (8), deriving the difference in acceptability. See Katzir (2024) for a more detailed discussion.

There are further aspects of the convertible examples that seem natural on the reduction of FF to QF but are surprising on the anaphoric account (see Katzir 2013, 2024). One can of course keep complicating the anaphoric (6) so as to accommodate these data, but this is no longer done for the empirical observations but in spite of them. The sole motivation to do so, as far as I can see, would be to avoid talking about reconstructed questions.

As it happens, however, the literature provides evidence that reconstructed questions are needed for an account of FF even if an anaphoric system is also adopted. Specifically, evidence from Spathas (2010) suggests that anaphoricity is not enforced at all nodes but at most in those corresponding to the reconstructed question, while Schwarzschild (1999) shows that minimality is relativized to this question.

Spathas's evidence concerns cases such as the following:

(9) (Sue talked to Bill's mother. No, ...) [MARY]$_F$ talked to #/✓Bob's mother

In many contexts, (9) requires accent on 'Bob', in line with the anaphoric account, since 'Bob' is not anaphoric and therefore requires F-marking. But Spathas notes that things change if Bill and Bob share a mother; in that case, not F-marking 'Bob' is the preferred option, even though 'Bob' is still not anaphoric in this context.[8] What seems to be the case is this: when Bill and Bob share a mother, call her x, the correction addresses the question of who talked to x. Elements that are within this reconstructed question (such as 'Bob') do not require $\sim$. But this requires taking into account the reconstructed question.[9]

Schwarzschild's evidence targets minimality and comes from F-marking in all-given contexts:

(10) (Was it a cheap convertible or an expensive one?) It was a [CHEAP]$_F$ convertible

Here 'cheap' is obligatorily F-marked even though the entire target sentence is given, presumably because it addresses the question of whether it was a cheap convertible or an expensive one. This suggests that minimality needs to be relativized to a reconstructed question.

In light of the above, the anaphoric (6) provides no advantages beyond (4) and (5).

3 Association with focus

(3) above illustrated AF: the possible entailments of a sentence with 'only' co-vary with focus placement. Moreover, these entailments can be stated in terms of the FSV of the prejacent of 'only' (specifically, 'only' can be said to negate certain elements of this FSV). Similar co-variation has been observed for other operators, including 'even', 'always', superlatives, and more.

A naive way to capture the entailments of sentences with operators such as 'only' is to say that these operators are lexically specified with reference to the FSV of their prejacent. Rooth (1985, 1992), however, cautions against such lexical specifications. If operators could have focus sensitivity written into their lexical entries, we would expect to find a great deal of variation in focus sensitivity across languages. For example, we might expect to find dialects of English where 'only' is not focus sensitive. In fact, though, we find no such variation, which suggests that AF arises from general principles rather than from arbitrary lexical specification.

Rooth (1992)'s derivation of AF is anaphoric, relying on the fact that operators quantify over restricted domains and using $\sim$ to communicate FSVs to these domains. If the domain of a given

[8] Without context, some oddness remains. As far as I can tell this is due to the unmotivated change of guise from Bill's mother to Bob's mother and has nothing to do with focus. If the change is motivated (e.g., if the speaker has reason not to want to utter Bill's name), the sentence improves. Either way, accenting 'Bob' only makes things worse.

[9] This is not Spathas (2010)'s own interpretation of his observation.

operator happens to be accidentally co-indexed with v of $\sim_v$, the operator will quantify over that value, a subset of the FSV of whatever $\sim_v$ attaches to. In a configuration such as [[Only D] [$\sim_v$ X], if D and v are co-indexed, 'only' will quantify over a subset of $[[\text{X}]]_f$. This makes 'only' focus sensitive simply by having D. Anaphoric AF is indirect (since the lexical entry of 'only' does not refer to focus), thus meeting Rooth's desideratum of deriving AF by general principle. It also has two further properties: it is structurally flexible, since co-indexation of variables is relatively unconstrained; and it is optional, since D and v are also free to not be co-indexed. Both structural flexibility and optionality are problematic.

Starting from structural flexibility, Krifka (2004) points out that AF operators such as 'only' seem incapable of using FSVs from constituents other than their prejacent. (11), for example, cannot mean that everyone who eats fruit and does not wear hats of any color says that Kim wears blue hats (perhaps in response to the question "What color hats does Kim wear?"):

(11) [Everyone who only$_D$ eats fruit] says that Kim [wears BLUE$_F$ hats]

AF using $\sim$, on the other hand, can co-index D and v even if the two are distant (similarly to the possible co-indexation in "[Everyone who knows him$_1$] says that [he$_1$ is smart]").

Anaphoric AF also incorrectly predicts optional AF for all operators with domain restriction, whereas the actual typology of AF is more complex. For some operators, such as 'always', focus sensitivity indeed seems optional, but for other operators, such as 'only', it appears to be obligatory. See Krifka 1992, von Fintel 1994, 1996, and especially Beaver and Clark 2008 for observations and discussion. The following is a descriptive approximation.

(12) Descriptive typology of focus sensitivity:
   a. Optional: 'always', 'usually', . . .
   b. Obligatory: 'only', 'even', . . .

What needs to be derived, then, is more nuanced than what $\sim$ predicts: structurally constrained AF, roughly based on the prejacent, and with the right kind of optionality or obligatoriness.

As with FF, I suggest that working directly from QF is not just more economical than the anaphoric approach but also more promising empirically. The idea is this: focus placement and domain of quantification co-vary the way they do since only certain pairings of focus placement and domain make a sentence a good answer to a good question. With a given focus placement in a sentence with an operator such as 'only' and 'always', speakers infer a domain that leads to a good pairing. This idea has already been explored by Roberts (1996) and is also very close in spirit to Schwarzschild (1993, 1997) and von Fintel (1994, 1996). Following is a brief outline of how this general idea applies to the descriptive typology in (12) (though the implementation details will differ from those in the works just mentioned).

Starting from (12a), the literature notes that the domain of operators such as 'always' can be set by inference (see, e.g., Schubert and Pelletier 1987), seemingly independently of focus.

(13) (Sue is very polite.) She **always**$_D$ says thank you

In (13), D may be understood as a salient set of times for which we might wonder whether Sue says thank you (presumably, when she receives something and is in a position to acknowledge it). Focus, which seems to fall broadly on [always$_D$ says thank you], seems unhelpful in reaching this D (as is presupposition, which is also sometimes invoked in domain restriction). Rather, (13) arguably involves a general inference to a salient D that makes it reasonable to wonder whether she says thank you for all times in D. If the sentence had been "She **always**$_{\mathrm{D}}$ says sorry", we would take D to be a different set of times (perhaps when Sue does something offensive and is in a position to apologize for it). Again, this can be explained as general inference to a set for which it makes sense to wonder whether she says sorry.

Focus can interact with this inferential resolution: D can be resolved to a salient set for which it makes sense to ask the question induced by the sentence. Consider the following:

(14) a. She **always**$_{\mathrm{D}}$ gave [COOKIES]$_{\mathrm{F}}$ to Bob
b. She **always**$_{\mathrm{D}}$ gave cookies to [BOB]$_{\mathrm{F}}$

In (14a), the question concerns the x such that at all times in D she gave x to Bob. If D is a salient set of times when she was in a position to give something to Bob, the question and the answer seem sensible. In (14b), the question concerns the x such that at all times in D she gave cookies to x. Here a choice of salient set of times when she was in a position to give cookies to someone seems sensible. The current view would of course need to say more about this inference.

Turning to (12b), obligatory AF operators (including 'only', 'even', 'too', etc.) seem to share the following property: they keep the question of their prejacent unchanged. (3a) (="She only gave [COOKIES]$_{\mathrm{F}}$ to Bob") is a possible answer to "What did she give to Bob?", and (3b) (="She only gave cookies to [BOB]$_{\mathrm{F}}$") is a possible answer to "Who did she give cookies to?".[10] So $Op_{\mathrm{D}}(\phi)$ answers the same question as ϕ. Pragmatically, this question is the partition Π of the context set C corresponding to some $Q \subseteq [[\ \phi\ ]]_{\mathrm{f}}$. To be relevant, $Op_{\mathrm{D}}(\phi) \cap C$ should be a union of cells in Π, and this constrains the possibilities for D. Consider again 'only$_{\mathrm{D}}(\phi)$', as in (3). 'Only' narrows down its prejacent ϕ, affirming ϕ and negating alternatives in D. $\phi \cap C$ is a union of cells in Π, so in order for the narrowing to maintain relevance, it must eliminate whole cells of Π in $\phi \cap C$. Negating elements of $[[\ \phi\ ]]_{\mathrm{f}}$, or any other proposition whose intersection with $\phi \cap C$ is a union of cells in Π, accomplishes this, but many other propositions do not. In (3a), negating the proposition that she gave cake to Bob maintains relevance but negating the proposition that she gave cookies to Bill violates it. In (3b), on the other hand, negating the proposition that she gave cake to Bob violates relevance while negating the proposition that she gave cookies to Bill maintains it.

The above comes close to the desideratum of obligatorily restricting D to Q, but it is not the same thing: the set of unions of cells in Π is a superset of Q, leading to at least one major worry. Suppose $Q = \{$that John saw Mary, that John saw Kim$\}$. On the standard view, "John only$_{\mathrm{D}}$ saw [Mary]$_{\mathrm{F}}$" can at most negate that John saw Kim, which is indeed a possible entailment of the sentence. But on the current view, D can include the cell that John saw Mary and did not see Kim, or the union of that cell with the cell that John saw neither Mary nor Kim. Negating either of these while affirming the prejacent entails that John saw Kim, an empirically impossible entailment of

[10] This poses an empirical challenge to congruence, at least as usually stated. See Schwarzschild (1997) for relevant discussion. In section 4 I will suggest that congruence should be dropped as a requirement and that its appearance should be derived from relevance. If correct, the challenge of sentences with 'only' answering the question of their prejacent despite not being congruent does not arise. Either way, of course, one would like to know why operators such as 'only' preserve the question of their prejacent.

the sentence. This is an instance of the so-called symmetry problem, and I don't know how to solve it within the current approach.[11]

4 Question focus

The anaphoric $\sim$ was unhelpful both for FF and AF. Its predictions often replicated those already made by a theory of QF, making the additional machinery involved in the anaphoric account of FF and AF unnecessary. And (setting aside the symmetry problem), when the predictions of $\sim$ did go beyond those of QF, they were wrong.

Should we keep $\sim$ for QF, to ensure congruence, as in (5)? I see no good reason to do so. All that $\sim$ did for congruence was enforce subsethood between the question that (4) refers to and the FSV of the sentence. But if that is what we wish to say, we might as well state it directly:

(15) CONGRUENCE ($\sim$-free):
  a. The denotation of the question that S addresses is a subset of $[\![\text{S}]\!]_{\text{f}}$
  b. S is minimally F-marked to achieve (15a)

We are left with no useful role for $\sim$ in the grammar, and with a range of serious challenges in both FF and AF if we do adopt it. I conclude that it does not exist.

More fundamentally, the above raises questions about the very idea of FSVs. If a second dimension of meaning such as $[\![\cdot]\!]_{\text{f}}$ exists, why is it only visible to something like (15)? Why, in particular, don't we find operators that can access FSVs or variables that can refer to them? Question-answer congruence seems to provide the only remaining empirical motivation for FSVs. This is not much, and note that statements such as (15) (or (5)) leave us no closer to understanding why congruence should hold in the first place. It would be better if we could eliminate FSVs from the grammar altogether and derive the appearance of congruence from more basic principles.

Here is what such a derivation might look like. F-marking a constituent X says that the choice of X matters.[12] If X is part of a larger ϕ, then it matters that $\phi[X]$ rather than $\phi[X']$ for some $X' \neq X$ (and possibly also for other replacements: X'', X''', etc.) — the choice between them is relevant. But recall that things are relevant in discourse only with respect to a pragmatic question, a partition Π of the context set C. Choices between different unions of cells in Π are relevant, while choices that cut across a cell are not. So indicating that the choice between X and X' within ϕ matters amounts to saying that $[\![\phi[X]]\!]_{\text{o}} \cap C$ and $[\![\phi[X']]\!]_{\text{o}} \cap C$ are different unions of cells in Π. This predicts a tight connection between F-marking and Π.[13]

Suppose, for example, that Π is about what she gave to Bob, with a cell for the worlds in which she gave cookies to Bob and nothing else, a cell for the worlds in which she gave cake to Bob and nothing else, and so on. This Π can arise from the explicit question "What did she give to

[11] Salience may perhaps seem promising (the elements of $[\![\phi]\!]_{\text{f}}$ are salient while unions of cells outside of $[\![\phi]\!]_{\text{f}}$ are not), but see Fox and Katzir (2011) for challenges to the idea that salience eliminates symmetry.

[12] I keep F-marking as a connection between prosody and pragmatics, but this is inessential. One may wish to connect the two more directly.

[13] In principle, the pragmatic account can be extended also to distinctions below the word level and distinctions of the kind that is sometimes referred to as meta-linguistic, phenomena that have proven hard to capture using FSVs (see Artstein 2004 and Wagner 2022). By connecting prosody and pragmatics, possibly with syntactic F-marking but without the semantic intermediary of FSVs, the current approach offers a potential handle on such cases, but I will not be able to explore this here.

Bob?”, as in (1a), but it can also be reconstructed in the absence of a semantic question. F-marking ‘cookies’ in the intended answer “She gave cookies to Bob” indicates a relevant choice: “She gave cookies to Bob” vs. “She gave X' to Bob’, for some $X' \neq$ cookies, each option being a union of cells within the question partition. Any other F-marking in the answer would violate relevance. For example, F-marking ‘Bob’ indicates an irrelevant choice: “She gave cookies to Bob” vs. “She gave cookies to Y'”, for some $Y' \neq$ Bob, a choice that cuts across cells in the partition. Suppose now that both ‘cookies’ and ‘Bob’ are F-marked. This indicates that both choices matter, ‘cookies’ vs. some other X' and ‘Bob’ vs. some other Y', and the latter choice again violates relevance. At least in this case, then, relevance requires F-marking to be minimal.[14] Differently from the more common view on QF, as stated in (5) and in (15), the current pragmatic view derives the appearance of congruence rather than stipulating it, and it makes no reference to FSVs.

The pragmatic derivation above suggests that FSVs, the second dimension of meaning commonly used for focus dependencies, can and should be eliminated. The derivation is not intended to replace a theory of alternatives, however: one still needs to explain which substitutions are used in which cases, which ones can be ignored, and so on. It is conceivable that an adequate theory of alternatives will be easier to state within a pragmatic derivation than within a grammatical theory of focus semantics, but it is also possible that the pragmatic setting will turn out to be incompatible with such a theory. I will have to leave this matter open here. The pragmatic approach also faces the challenge of accounting for embedded effects of focus sensitivity. I have no solution for this, either.

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[14]For a more general derivation of minimality, one needs to prevent the possibility of ignoring replacements that are irrelevant given the background pragmatic question. I will have to set this aside here.

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